

LSSO: Solving Contextual Adaptation with Certified Global Mixing

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Abstract

Context-conditioned sequence layers commonly express adaptation through finite update rules or recurrent state evolution. We introduce LSSO, a global mixer that instead turns contextual adaptation into a solved operator: one full-sample statistic generates a compact strictly accretive system, and a direct solve returns its equilibrium. A QR soft frame lifts the compact operator to token space, while a learned scalar complement controls the unrepresented subspace. We prove that the compact parameterization covers the entire open real matrix ball and that every frozen, sample-conditioned token mixer is a strict ℓ_2 contraction whose deviation from the scalar complement has rank at most r . The returned state is the exact infinite-time limit of a globally convergent contextual flow, without unrolling, an inner learning rate, or a test-time gradient update. We also provide a complete native CUDA path for inference and first-order training of the default operator, including an analytic backward that reuses the compact factors. In a matched eight-task GenomicBenchmarks comparison, LSSO is higher than multi-head attention on seven tasks and improves unweighted macro accuracy from 75.37 to 78.53. Four Long Range Arena tasks test the same operator contract on sequences of up to 4096 positions.

1 Introduction

Global interaction is central to sequence and vision encoders, yet materializing all token pairs incurs quadratic cost. Efficient alternatives compress the sequence into landmarks or latent slots, replace attention by structured recurrence, or maintain an input-selective state [24, 12, 15, 9, 6, 5, 3]. A second line of work interprets sequence processing itself as learning: fast-weight models and test-time training adapt a compact inner state to the current context [19, 20]. Together these developments suggest that the central object need not be a dense attention matrix; it can be a small state constructed from the sample.

The remaining question is how that state should be obtained. Finite inner optimization introduces an objective, step size, update schedule, and unrolling depth. ViT³ shows that these choices materially shape visual test-time training, including the optimization of deeper inner models [8]; more complete inner solves can also require substantial inference work [22]. Equilibrium networks offer well-posed implicit states through monotone structure [25], but usually solve a static learned operator iteratively. We study the intersection of these ideas: a compact operator generated by the current sample, with an equilibrium that is available in closed form and a geometry that can be verified for every generated instance.

This leads to our central question:

Can contextual adaptation be solved rather than unrolled, without giving up global conditioning, linear sequence scaling, or a verifiable operator geometry?

LSSO answers this question with a compact matrix-ball construction. Each head compresses the full sample into a shared cross-moment $Z = P^\top C$, which generates a strictly accretive matrix K . Its contextual state is returned by the direct solve $U^\star = (\mathbf{I} + K)^{-1}Z$. Equivalently, the compact mixer is the reflected resolvent

$$M = 2(\mathbf{I} + K)^{-1} - \mathbf{I} \quad (1)$$

of the generated accretive system. A QR soft frame P and a learned scalar complement η lift this state to tokens,

$$T = \eta\mathbf{I} + P(M - \eta\mathbf{I})P^\top. \quad (2)$$

The construction never materializes T . It evaluates token-rank products and one small solve, giving linear sequence scaling for fixed rank. The accretive chart is complete, its reflected resolvent spans every real strict

compact contraction, and the scalar complement controls rather than hard-codes the token subspace outside P .

LSSO therefore occupies a previously disconnected point in the design space: it is a noncausal global compact-state mixer, a solved form of contextual adaptation, and a per-instance certified operator. Here and throughout, “certified” refers to the realized token mixer after its sample-conditioned quantities are fixed, not to the end-to-end Jacobian of the surrounding network. The same algebra also admits a direct hardware realization. Our default DYNAMIC plus Rank-Rotary operator has native CUDA inference, training forward, and analytic first-order backward paths. Compact construction and factorization share one schedule, the backward reuses saved factors, and neither an $N \times N$ token map nor an explicit compact Cayley matrix is formed.

Our contributions are fourfold:

1. **Solved adaptation without unrolling.** We introduce a noncausal global mixer whose sample-conditioned inner state is the exact solution of a compact accretive system. At fixed rank it scales linearly with sequence length and removes inner learning rates, update counts, and fixed-point iterations from the forward pass.
2. **A complete contraction parameterization.** We show that the compact construction spans the open real matrix ball and that defect completion lifts it to a strictly contractive token mixer with a rank- r correction around a learned scalar complement. The direct solution is also the infinite-time limit of a globally convergent contextual flow.
3. **An analytic, hardware-efficient realization.** We lower the solved operator and its analytic backward to native CUDA without forming a token–token map. The resulting complete mixer is up to $2.30\times$ as fast as Flash-SDPA MHA on the measured long-sequence workloads, while its outputs and parameter gradients remain within a fixed FP64-oracle envelope.
4. **Evidence on long structured inputs.** In a matched three-seed comparison across eight genomic tasks, LSSO improves over MHA on seven tasks and raises macro accuracy by 3.16 points. A four-task LRA panel further spans symbolic, byte-level, retrieval, and spatial sequences up to length 4096.

2 Related Work

Global compact-state mixing. Efficient Transformers replace the dense token map with projections, landmarks, kernels, or inducing states [24, 10, 1, 26]. Set Transformer uses inducing points for global set interaction [12]; Luna packs and unpacks information through a fixed-length auxiliary sequence [15]; and ABC interprets several such mechanisms as learned bounded memory [18]. Perceiver similarly mediates global exchange through latent slots [9]. LSSO advances this compact-state line in a different direction: the sample generates a compact operator once per head, a direct solve returns its certified accretive equilibrium, and the resulting global mixer is applied without constructing a dense token map.

Context as learning. Fast-weight views interpret linear attention as programming an associative memory [19]. TTT learns an inner model from the current sequence [20], while Longhorn and test-time regression connect state updates to online least-squares objectives [14, 23]. ViT³ exposes the design and optimization of inner models as a central axis in vision [8]; MesaNet moves toward more complete per-context optimization with a parallel iterative solver [22]. LSSO realizes a solved endpoint of this perspective for bidirectional encoders: the full context constructs one linear system, and its exact equilibrium replaces a finite update schedule.

Structured sequence models and certified equilibria. S4 and selective state-space models obtain linear sequence scaling through structured recurrence and input-dependent state evolution [6, 5, 3]. Monotone operator equilibrium networks obtain well-posed implicit states from monotone structure [25]. LSSO combines global noncausal conditioning with the corresponding accretive/resolvent geometry: each generated compact system has a unique equilibrium, and defect completion transfers its contraction certificate to the realized

token kernel. Ordinary self-attention need not be globally Lipschitz [11]; Section 7 distinguishes our per-kernel theorem from an end-to-end network guarantee.

Hardware-efficient linear layers. FlashAttention demonstrated that avoiding materialized intermediates and controlling memory traffic can matter as much as operation count [2]; Mamba likewise couples its recurrence to a hardware-aware parallel algorithm [5]. LSSO follows this algorithm–implementation co-design principle. Its native path lowers the compact algebra to fused CUDA schedules for both the forward and analytic backward, while retaining FP32 factorization and solves where the numerical contract requires them. Section 6 summarizes the execution path, and Appendix B gives the complete lowering and workspace accounting.

3 LSSO

We describe one head and omit the batch index. Let $X \in \mathbb{R}^{N \times D}$ be the incoming token matrix, d the head width, and r the compact rank. A single input projection produces relation and content coordinates

$$[A, C] = XW_{bc} + b_{bc}, \quad A \in \mathbb{R}^{N \times r}, \quad C \in \mathbb{R}^{N \times d}. \quad (3)$$

Heads are concatenated and passed through an ordinary output projection after mixing. The construction below is repeated independently for each head, except that the same head’s compact state couples all of its valid tokens.

3.1 Masking and rank-space phase

For a boolean validity mask $m \in \{0, 1\}^N$, invalid rows of both A and C are set to zero before any compact statistic is formed. Let

$$n = \sum_{i=1}^N m_i. \quad (4)$$

For a nonempty sequence, relation coordinates are normalized as $A \leftarrow A/\sqrt{n}$. An entirely masked sequence is assigned the zero output by definition. This convention makes padding a semantic input rather than a separate dense-only path.

The optional Rank-Rotary ablation applies a block-diagonal planar rotation to the r relation coordinates before the soft frame. For centered position p_i and frequencies $\{\omega_j\}_{j=1}^{r/2}$, each coordinate pair uses the angle $p_i\omega_j$. The transformation preserves each relation row’s Euclidean norm and obeys the relative-phase identity

$$\mathcal{R}(p_i)^\top \mathcal{R}(p_j) = \mathcal{R}(p_j - p_i). \quad (5)$$

It is therefore an internal phase choice in rank-feature space. It is not a RoPE replacement, an absolute-position representation, or a claim of two-dimensional spatial equivariance.

3.2 QR soft frame and compact statistic

The QR soft frame maps unconstrained relation coordinates to a strict column contraction. Form the reduced QR factorization

$$\begin{bmatrix} P \\ E \end{bmatrix} R_{\text{qr}} = \begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix}, \quad \begin{bmatrix} P \\ E \end{bmatrix}^\top \begin{bmatrix} P \\ E \end{bmatrix} = \mathbf{I}_r, \quad (6)$$

where $P \in \mathbb{R}^{N \times r}$ is the token-space frame. The implementation applies a detached positive common scale before this QR; it leaves the exact Q factor unchanged and keeps the numerical evaluation in a normal range. The identity augmentation means that the factorization is defined even when $N < r$ or A is rank deficient.

The frame compresses the content into one shared cross-moment,

$$Z = P^\top C \in \mathbb{R}^{r \times d}. \quad (7)$$

Unlike a recurrent state, Z summarizes all valid positions symmetrically in one noncausal pass. The token-level matrix in (2) is conceptual only; it is never instantiated.

3.3 Accretive compact equilibrium

In the default DYNAMIC mode, Z generates compact raw coordinates

$$\Theta = \Theta_0 + \frac{ZW_{\text{drive}}}{\sqrt{n}}, \quad (8)$$

where $\Theta_0 \in \mathbb{R}^{r \times r}$ and $W_{\text{drive}} \in \mathbb{R}^{d \times r}$ are learned per-head parameters. The STATIC ablation uses $\Theta = \Theta_0$, and ZERO owns no compact coordinates. No other core variants are part of the operator contract.

For DYNAMIC and STATIC, define

$$L(\Theta) = \text{tril}(\Theta, -1) + \text{Diag}(\text{softplus}(\text{diag}(\Theta) + c_1)), \quad (9)$$

$$\Omega(\Theta) = \text{triu}(\Theta, 1) - \text{triu}(\Theta, 1)^\top, \quad (10)$$

$$K(\Theta) = L(\Theta)L(\Theta)^\top + \Omega(\Theta), \quad (11)$$

where $c_1 = \text{softplus}^{-1}(1)$. Thus L is lower triangular with positive diagonal, while Ω is skew-symmetric. The compact equilibrium and the pre-output token coordinates are

$$U^\star = (\mathbf{I}_r + K)^{-1}Z, \quad (12)$$

$$Y = \eta C + P[2U^\star - (1 + \eta)Z], \quad \eta = \tanh(\eta_{\text{raw}}) \in (-1, 1). \quad (13)$$

This is the exact-real map. The floating-point implementation evaluates a stable logistic identity and multiplies by one minus the calculation-dtype machine epsilon, reserving one ULP from $\{-1, 1\}$. The complement is learned independently for each head and is initialized nominally to 0.9. In ZERO, we set the compact term to $M = 0$, which gives

$$Y = \eta(C - PZ). \quad (14)$$

At $\Theta = 0$, $L = \mathbf{I}_r$, $\Omega = 0$, $K = \mathbf{I}_r$, and the compact contraction is $M = 0$. Both nonzero-core modes start at that point; DYNAMIC also starts with $W_{\text{drive}} = 0$.

For later use, define the reflected resolvent

$$M = 2(\mathbf{I}_r + K)^{-1} - \mathbf{I}_r = (\mathbf{I}_r - K)(\mathbf{I}_r + K)^{-1}. \quad (15)$$

Substituting $Z = P^\top C$ into (13) gives the exact token-mixing form

$$Y = TC, \quad T = \eta \mathbf{I}_N + P(M - \eta \mathbf{I}_r)P^\top. \quad (16)$$

Solved contextual inner state. Equation (13) is the exact solution of a small linear system. Section 4.3 proves that it is the infinite-time limit of a globally convergent contextual flow and, separately, the infinite-step limit of gradient descent on a residual objective. These are mathematical interpretations, not the computation performed by LSSO. In particular, when $\Omega \neq 0$, the original equation $(\mathbf{I}_r + K)U = Z$ is not the first-order condition of a ridge or other symmetric convex energy in U .

Cost. For fixed r and d , frame formation costs $O(Nr^2)$ per head, the frame-content products cost $O(Nrd)$, and the compact solve costs $O(r^2d + r^3)$. The mixer is therefore linear in N at fixed compact dimensions and never stores an $N \times N$ token matrix. The dense input and output projections remain ordinary feature projections, as in other token mixers.

4 Solved and Certified Contextual Mixing

All statements in this section are exact-real-arithmetic statements about a single head after its input-conditioned quantities have been generated. The mixed-precision numerical contract is discussed separately in Section 6.

4.1 The QR soft frame is a strict contraction chart

Theorem 4.1 (Soft-frame contract). *Let P be the upper block of the Q factor in (6). Then*

$$P^\top P \prec \mathbf{I}_r, \quad \mathbf{I}_N - PP^\top \succ 0. \quad (17)$$

Moreover, at the level of represented frames, the construction covers every $P \in \mathbb{R}^{N \times r}$ satisfying $P^\top P \prec \mathbf{I}_r$.

Proof sketch. The lower block of (6) obeys $ER_{\text{qr}} = \mathbf{I}_r$. Since the augmented matrix has full column rank, R_{qr} and hence E are nonsingular. Orthogonality of the full thin- Q factor gives

$$P^\top P + E^\top E = \mathbf{I}_r. \quad (18)$$

Because $E^\top E \succ 0$, the first claim follows. In particular, $\|P\|_2 < 1$, which implies $PP^\top \prec \mathbf{I}_N$ and proves the second claim. Appendix A constructs the inverse chart for every P in the strict column ball. $\square$

The identity augmentation is more than a numerical safeguard. It supplies a strict defect $\mathbf{I}_N - PP^\top$, which will let us complete the compact contraction to token space without an uncontrolled residual subspace.

4.2 The compact core parameterizes the strict matrix ball

Definition 4.2 (Strict accretivity). *A real square matrix K is strictly accretive when*

$$\text{sym}(K) := \frac{K + K^\top}{2} \succ 0. \quad (19)$$

Theorem 4.3 (Accretive Cayley parameterization). *The map in (11) ranges over all real strictly accretive $r \times r$ matrices. For every such K , the reflected resolvent M in (15) is a strict spectral-norm contraction and obeys*

$$\mathbf{I}_r - M^\top M = 4(\mathbf{I}_r + K)^{-\top} LL^\top (\mathbf{I}_r + K)^{-1} \succ 0. \quad (20)$$

Conversely, the map $K \mapsto M$ is a bijection from real strictly accretive matrices to the open real matrix ball $\{M : \|M\|_2 < 1\}$, with inverse

$$K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}. \quad (21)$$

Proof sketch. The construction has $\text{sym}(K) = LL^\top \succ 0$, because L has positive diagonal. Conversely, for any strictly accretive K , the Cholesky factor of $\text{sym}(K)$ and the skew remainder $K - \text{sym}(K)$ recover L and Ω , proving completeness.

Let $Q = \mathbf{I}_r + K$. Strict accretivity implies that Q is nonsingular. Using $M = (\mathbf{I}_r - K)Q^{-1}$ gives

$$\mathbf{I}_r - M^\top M = Q^{-\top} [Q^\top Q - (\mathbf{I}_r - K)^\top (\mathbf{I}_r - K)] Q^{-1} \quad (22)$$

$$= 4Q^{-\top} \text{sym}(K) Q^{-1}, \quad (23)$$

which is (20). It is positive definite, so $\|M\|_2 < 1$. The inverse formula maps every strict contraction back to a strictly accretive matrix. Appendix A gives the complete converse and fractional-linear algebra. $\square$

Corollary 4.4 (Closed-form strongly monotone equilibrium). *For every $Z \in \mathbb{R}^{r \times d}$, the equilibrium $U^* = (\mathbf{I}_r + K)^{-1}Z$ exists uniquely. Moreover,*

$$\sigma_{\min}(\mathbf{I}_r + K) \geq 1. \quad (24)$$

The proof is given in Appendix A.

4.3 The direct solve is an infinite inner-process limit

Let $A_K = \mathbf{I}_r + K$. The following result states precisely in what sense the forward pass replaces an infinite contextual inner process. It does not identify LSSO with an arbitrary TTT objective or nonlinear inner model.

Theorem 4.5 (Solved contextual inner process). *For fixed contextual quantities K and Z , consider the continuous dynamics*

$$\dot{U}(t) = Z - A_K U(t). \quad (25)$$

From every initial state $U(0)$, the dynamics converge to $U^ = A_K^{-1}Z$ and obey*

$$\|U(t) - U^*\|_F \leq e^{-t} \|U(0) - U^*\|_F. \quad (26)$$

The rate lower bound is independent of the skew-symmetric part of K .

Proof sketch. Write $E(t) = U(t) - U^*$. Since $A_K U^* = Z$, $\dot{E} = -A_K E$. The skew part vanishes in the Frobenius inner product, so

$$\frac{d}{dt} \frac{1}{2} \|E\|_F^2 = -\langle E, A_K E \rangle_F = -\langle E, \text{sym}(A_K) E \rangle_F \leq -\|E\|_F^2, \quad (27)$$

because $\text{sym}(A_K) = \mathbf{I}_r + LL^\top \succeq \mathbf{I}_r$. Gronwall's inequality gives (26). $\square$

Appendix A further shows that this direct solution is the infinite-step limit of stable gradient descent on the residual objective and derives its exact implicit VJP and sensitivity bound. Thus the production backward differentiates the converged state rather than a finite unroll.

4.4 Defect completion lifts the certificate to tokens

Theorem 4.6 (Conditional contraction and low-rank structure). *Fix a sample and head, and let P , M , and η be constructed as above. Then the token mixer in (16) satisfies*

$$\|T\|_2 < 1, \quad \text{rank}(T - \eta \mathbf{I}_N) \leq r. \quad (28)$$

For every $v \in \ker(P^\top)$, it further obeys $Tv = \eta v$.

Proof sketch. By Theorem 4.1, $\Delta = (\mathbf{I}_N - PP^\top)^{1/2}$ is well-defined and

$$F = [P \ \Delta], \quad FF^\top = \mathbf{I}_N. \quad (29)$$

The token mixer admits the defect-completed factorization

$$T = PMP^\top + \eta(\mathbf{I}_N - PP^\top) = F \begin{bmatrix} M & 0 \\ 0 & \eta \mathbf{I}_N \end{bmatrix} F^\top. \quad (30)$$

Theorem 4.3 gives $\|M\|_2 < 1$, while $\eta = \tanh(\eta_{\text{raw}})$ gives $|\eta| < 1$. Since F is a coisometry,

$$\|T\|_2 \leq \max\{\|M\|_2, |\eta|\} < 1. \quad (31)$$

Finally,

$$T - \eta \mathbf{I}_N = P(M - \eta \mathbf{I}_r)P^\top, \quad (32)$$

which has rank at most r . If $P^\top v = 0$, the same expression immediately gives $Tv = \eta v$. $\square$

Remark 4.7 (Why the scalar complement is part of the model). *Without the complement, a conventional form $\mathbf{I}_N + PAP^\top$ fixes the entire $\ker(P^\top)$ subspace to gain one. Equation (30) instead assigns that unmodeled subspace the learned per-head scalar η , while retaining the same rank- r correction and the strict contraction certificate. This does not make T an arbitrary full-rank attention map: outside the represented subspace it remains isotropic.*

Padding isolation and noncausal permutation equivariance follow from the same rowwise construction; Appendix A states and proves both properties. The precise scope of the frozen-kernel certificate is collected in Section 7.

5 Sequence Classification Evidence

We first test whether the certified mixer remains trainable and useful when it replaces attention in otherwise ordinary bidirectional encoders. All reported LSSO runs use the default DYNAMIC core, Rank-Rotary, rank 32, FP16 automatic mixed precision, and the CUDA implementation. Checkpoints are selected only by validation accuracy, with validation loss breaking an exact tie; the held-out test split is evaluated once after selection. Reported uncertainties are sample standard deviations over seeds 0, 1, and 2.

5.1 Matched genomic mixer comparison

GenomicBenchmarks collects binary DNA sequence-classification tasks with official train and test partitions [4]. We derive a deterministic stratified validation set only from each training partition. MHA and LSSO use the same $D = 128$, four-layer, four-head encoder, learned absolute position embeddings, MLP ratio 4, mean readout, optimizer, data order, and 40-epoch budget; only the token mixer changes. The usual physical batch is 128. Mouse Enhancers Ensembl uses physical batch 64 and two-step accumulation, preserving the same effective batch. LSSO has 33,264 fewer parameters than MHA in every task because the mixer parameterizations differ; for the 200-position tasks the totals are 786,834 and 820,098.

Table 1: GenomicBenchmarks held-out test accuracy (%), mean $\pm$ sample standard deviation over three seeds. Δ is LSSO minus MHA. This is a matched mixer comparison, not a state-of-the-art claim.

Task	MHA	LSSO	Δ
Coding vs. intergenomic	80.34 ± 1.99	83.23 ± 2.46	+2.89
Human or worm	65.87 ± 0.35	81.57 ± 1.18	+15.70
Mouse enhancers Ensembl	71.07 ± 1.65	70.25 ± 0.83	-0.83
Human enhancers Cohn	66.83 ± 0.17	67.24 ± 0.26	+0.41
Human enhancers Ensembl	76.37 ± 0.20	78.91 ± 0.05	+2.54
Human Ensembl regulatory	92.14 ± 0.05	93.10 ± 0.02	+0.96
Human non-TATA promoters	84.71 ± 0.15	85.41 ± 0.17	+0.70
Human OCR Ensembl	65.61 ± 0.54	68.55 ± 0.31	+2.93
Unweighted task macro	75.37	78.53	+3.16

Table 1 shows that LSSO is higher on seven of eight tasks. The macro result first averages seeds within each task and then gives all eight tasks equal weight. The largest gain occurs on Human or Worm, while Mouse Enhancers Ensembl is the one negative result. With only three seeds, we do not attach per-task significance claims to these differences.

5.2 Long Range Arena context

Long Range Arena (LRA) tests structured reasoning and long-context classification [21]. Its reported scores are unusually sensitive to architectural locality: recent analysis shows that explicit local and positional structure can account for a substantial fraction of the benchmark gains [17]. We therefore separate global token mixers without an explicit locality or recurrent-state module from models that add a structured state-space or locality prior. S4 and related models remain important LRA references, but their recurrence/locality prior tests a broader architectural package than global mixing alone. The split does not assert that either block is free of other inductive biases or position encodings; LSSO itself uses learned absolute positions in these shells.

We report four completed LSSO tasks. The shell is selected per task: ListOps uses $D = 128, L = 6$; Text uses $D = 256, L = 6$; Retrieval uses $D = 128, L = 6$; and Pathfinder-32 uses $D = 256, L = 6$ with factorized learned row/column position embeddings, no PEG, and a mean-maximum readout. The first three use learned one-dimensional absolute position embeddings and mean readout. Maximum token lengths are 2048, 4096, 4000, and 1024, respectively.

Table 2: LRA test accuracy (%). Literature values are copied from the cited papers and were not reproduced here. The first block contains global token mixers without an explicit locality or recurrent-state module; the second adds an explicit structured state-space or locality prior. Luna uses five seeds and a longer Retrieval budget; FNet uses its authors’ independent implementation; Nyströmformer uses its independent PyTorch protocol and a depthwise-convolution value skip. LSSO is mean $\pm$ sample standard deviation over three seeds under our task-specific protocol.

Model and source	ListOps	Text	Retrieval	Pathfinder
<i>Global token mixers without explicit locality or recurrent state</i>				
Transformer [21]	36.37	64.27	57.46	71.40
Linear Transformer [21]	16.13	65.90	53.09	75.30
Performer [21]	18.01	65.40	53.82	77.05
Linformer [21]	35.70	53.94	52.27	76.34
Reformer [21]	37.27	56.10	53.40	68.50
Luna-128 [15]	38.01	65.74	79.55	78.89
FNet [13]	35.33	65.11	59.61	77.80
LSSO, three seeds	37.60 $\pm$ 0.65	65.19 $\pm$ 0.46	82.88 $\pm$ 0.08	78.79 $\pm$ 0.34
<i>Models with explicit structured state-space or locality priors</i>				
Nyströmformer [26]	37.15	65.52	79.56	70.94
S4, original [6]	58.35	76.02	87.09	86.05
S4-LegS, updated [7]	59.60 $\pm$ 0.07	86.82 $\pm$ 0.13	90.90 $\pm$ 0.15	94.20 $\pm$ 0.25
Mega [16]	63.14	90.43	91.25	96.01

Table 2 places LSSO beside a broader set of global token mixers and separately identifies the structurally augmented references. Across the first group, LSSO is competitive on ListOps, Text, and Pathfinder and gives the strongest reported Retrieval value in the displayed panel. The broader result is that one operator contract trains across symbolic, byte-level, paired-retrieval, and two-dimensional path tasks, including sequences far longer than the genomic panel. Cross-paper recipes still differ, so the table supplies literature context rather than a controlled ranking. Path-X and an exploratory Pathfinder run with Grid-PEG are excluded because they are not part of the completed three-seed protocol.

6 Native CUDA Realization

The default DYNAMIC plus Rank-Rotary operator has a native, algebraically equivalent CUDA path for inference and first-order training. It never forms an $N \times N$ token map or the compact Cayley matrix; compact factorization and tiled solves evaluate the equilibrium directly, while the analytic backward reuses saved factors. Mixed-precision outputs and gradients are validated against the FP64 reference. Appendix B gives the execution schedule, workspace, VJP, numerical envelope, and build coverage.

Table 3 measures the complete mixer boundary, including input and output projections. Native LSSO is faster than the canonical PyTorch implementation throughout the reported range. Against PyTorch MHA backed by Flash SDPA, it crosses over by $N = 2048$ in the forward and by $N = 1024$ in forward-backward, reaching $1.79\times$ and $2.30\times$ speedups at $N = 4096$. These are single-device measurements, not a cross-architecture claim.

7 Limitations

The contraction theorem concerns the realized token kernel after its sample-conditioned frame and compact core are fixed. The full map $X \mapsto T(X)C(X)$ also differentiates through those generated quantities, the feature projections, residual connection, and surrounding MLP, so the result is not an end-to-end Lipschitz certificate for the complete network. The rank- r correction around a scalar complement is also a deliberate structural restriction rather than a parameterization of every dense attention map. Finally, LSSO is a noncausal global encoder mixer and does not provide constant-state autoregressive decoding.

Table 3: Synchronized milliseconds per batch on an RTX 5070 Ti (lower is better), with $B = 64, D = 256, H = 8$, FP16 AMP, and $r = 32$ for LSSO. Ratios above one favor native LSSO; forward-backward includes loss and gradient reset.

N	PyTorch ref.	Native	MHA	Ref./native	MHA/native
<i>Inference forward</i>					
1024	110.216	2.539	2.131	$43.41\times$	$0.84\times$
2048	133.940	5.577	5.888	$24.02\times$	$1.06\times$
4096	169.549	10.135	18.173	$16.73\times$	$1.79\times$
<i>Forward-backward with gradient reset</i>					
1024	118.858	7.512	8.487	$15.82\times$	$1.13\times$
2048	153.955	15.260	22.940	$10.09\times$	$1.50\times$
4096	211.268	29.160	67.195	$7.25\times$	$2.30\times$

The experiments establish a matched genomic mixer comparison and cross-modality trainability on four LRA tasks. LRA rows taken from the literature use different model shells and training protocols, and should be read as context rather than a controlled leaderboard. Controlled ablations of the three core modes and Rank-Rotary, cross-architecture wall-time and peak-memory sweeps, and the planned ImageNet, COCO, and ADE20K evaluations remain necessary before making broader claims about efficiency or vision backbones.

8 Conclusion

LSSO turns contextual adaptation from a finite update procedure into a solved global operator. One full-sample statistic generates a compact strictly accretive system, a direct solve returns its equilibrium, and defect completion lifts the resulting strict contraction to token space. The construction has linear sequence scaling at fixed rank and a complete native CUDA forward and analytic backward realization. GenomicBenchmarks and LRA results show that the same certified operator remains useful across long biological, symbolic, textual, retrieval, and spatial inputs.

A Complete Theoretical Details

A.1 Coverage of the QR soft-frame chart

Take any $P \in \mathbb{R}^{N \times r}$ with $P^\top P \prec \mathbf{I}_r$ and write $S = \mathbf{I}_r - P^\top P \succ 0$. Let C be the upper-triangular Cholesky factor of S , so $S = C^\top C$, and set $R_{\text{qr}} = C^{-1}$ and $A = PR_{\text{qr}}$. Then

$$\begin{bmatrix} A \\ \mathbf{I}_r \end{bmatrix} = \begin{bmatrix} P \\ R_{\text{qr}}^{-1} \end{bmatrix} R_{\text{qr}}. \quad (33)$$

The left factor has orthonormal columns because $R_{\text{qr}}^{-\top} R_{\text{qr}}^{-1} = S$. It is therefore the unique positive-diagonal reduced QR factorization and produces the desired upper block P .

A.2 Converse Cayley map and solve bound

For any M with $\|M\|_2 < 1$, the matrix $\mathbf{I}_r + M$ is nonsingular. Define $K = (\mathbf{I}_r - M)(\mathbf{I}_r + M)^{-1}$. Direct substitution gives

$$K + K^\top = 2(\mathbf{I}_r + M)^{-\top}(\mathbf{I}_r - M^\top M)(\mathbf{I}_r + M)^{-1} \succ 0. \quad (34)$$

Thus K is strictly accretive, and the forward and inverse fractional-linear expressions are algebraic inverses wherever defined.

For a nonzero v , the skew part of K contributes zero to the quadratic form, so

$$\langle (\mathbf{I}_r + K)v, v \rangle = \|v\|_2^2 + \|L^\top v\|_2^2 \geq \|v\|_2^2. \quad (35)$$

Cauchy-Schwarz therefore gives $\|(\mathbf{I}_r + K)v\|_2 \geq \|v\|_2$, proving $\sigma_{\min}(\mathbf{I}_r + K) \geq 1$ and $\|(\mathbf{I}_r + K)^{-1}\|_2 \leq 1$.

A.3 Residual optimization, implicit derivative, and sensitivity

The solved state also uniquely minimizes

$$J(U) = \frac{1}{2} \|A_K U - Z\|_F^2, \quad A_K = \mathbf{I}_r + K, \quad (36)$$

whose Hessian is $A_K^\top A_K \succ 0$. Gradient descent with $0 < \alpha < 2/\sigma_{\max}(A_K)^2$ obeys

$$\|U_j - U^*\|_F \leq \rho^j \|U_0 - U^*\|_F, \quad \rho = \max_i |1 - \alpha \sigma_i(A_K)^2| < 1. \quad (37)$$

Hence the direct solve is both the infinite-time limit in Theorem 4.5 and the infinite-step limit of every such residual-gradient run.

Differentiating $A_K U^* = Z$ gives

$$dU^* = A_K^{-1}(dZ - dA_K U^*). \quad (38)$$

If $G = \partial \mathcal{L} / \partial U^*$ and $V = A_K^{-\top} G$, trace cyclicity yields the exact implicit VJP

$$\frac{\partial \mathcal{L}}{\partial Z} = V, \quad \frac{\partial \mathcal{L}}{\partial A_K} = -V U^{*\top}. \quad (39)$$

The solve bound above further gives

$$\|U^*\|_F \leq \|Z\|_F, \quad \|dU^*\|_F \leq \|dZ\|_F + \|dA_K\|_2 \|Z\|_F. \quad (40)$$

A.4 Padding isolation and permutation equivariance

If an invalid row has $A_i = C_i = 0$, the upper QR block has $P_i = 0$, so Equation (13) gives $Y_i = 0$. Explicit output masking preserves this value after the feature projection.

Let Π permute token rows together with their mask and position identifiers. All rowwise projections, masking, normalization, and phase rotations commute with Π . Uniqueness of the positive-diagonal reduced QR factorization gives $P' = \Pi P$, while $Z' = P'^\top C' = P^\top C = Z$. The generated K , M , and η are unchanged, and Equation (16) yields

$$\text{LSSO}(\Pi X, \Pi m, \Pi p) = \Pi \text{LSSO}(X, m, p). \quad (41)$$

B CUDA Implementation Details

B.1 Execution contract

The native backend implements the complete default operator, DYNAMIC with Rank-Rotary, for inference and first-order training. It is an alternative lowering of the equations in Section 3 rather than a separate mathematical definition. The STATIC, ZERO, and Rank-Rotary-off ablations remain reference-only and are rejected at the native boundary. Unsupported inputs never silently fall back to another implementation, and higher-order gradients are outside the native contract.

The public CUDA entry point accepts FP16 or FP32 activations, while learned parameters remain FP32. The joint input projection in Equation (3) is evaluated outside the native mixer with FP32 operands and output, with TF32 enabled on supporting hardware. Its contiguous FP32 relation and content coordinates are passed to the native operator. The ordinary output projection is also outside the native boundary; it uses FP16 Tensor-Core operands with FP32 accumulation before the result is cast back to the activation dtype. The implementation therefore covers the complete nonlinear mixer and its first-order derivative without claiming that the surrounding feature projections form one monolithic kernel.

The schedule supports

$$r \in \{16, 32, 48, 64\} \quad (42)$$

and any positive head width d within ordinary allocation and launch limits. Rank and head width are independent, including $r > d$. Runtime right-hand-side panels have width 32 with zero-filled tails, so there is no second whitelist for d . BF16 is not part of this numerical contract.

Boolean validity masks have shape $[B, N]$, and position identifiers have shape $[N]$ or $[B, N]$. Invalid projected rows are zeroed before every compact statistic, all sample-dependent normalizations use $\sqrt{n_{\text{valid}}}$, and invalid output rows are zeroed again. This covers gapped padding and entirely masked samples in one generic schedule. Position identifiers and valid-token counts are metadata rather than differentiable inputs.

B.2 Lowering to tiled kernels

The forward pass begins with a rank-phase schedule that constructs the Rank-Rotary sine and cosine table. Consecutive default positions are cached by device, rank, and sequence length; explicit positions use an invocation-local table. One relation-input kernel combines phase rotation, $1/\sqrt{n_{\text{valid}}}$ normalization, detached common scaling, and storage of the scaled relation coordinates.

The soft frame uses the Gram–Cholesky form algebraically equivalent to the positive-diagonal reduced QR in Equation (6). Short sequences use a single-block FP32 Gram reduction. At 32 or more 32-token tiles, independent partial Grams are reduced in ascending token order before the same Cholesky factorization. Tiled triangular solves then materialize the frame. The cross-moment $Z = P^T C$ is produced in independent token and right-hand-side tiles using FP16 Tensor-Core multiplicands and FP32 accumulation; no more than 64 token partials are live at once.

A compact-core kernel evaluates

$$\Theta = \Theta_0 + \frac{ZW_{\text{drive}}}{\sqrt{n_{\text{valid}}}}, \quad (43)$$

constructs L and Ω , forms LL^T using IEEE FP32 FMA, assembles $\mathbf{I} + K$, and performs partial-pivot LU factorization. Separate 32-column FP32 solve panels compute $U^* = (\mathbf{I} + K)^{-1}Z$. The final tiled kernel directly evaluates

$$Y = \eta C + P[2U^* - (1 + \eta)Z]. \quad (44)$$

Thus neither the Cayley matrix M nor a token-space transition matrix is stored.

B.3 Analytic backward and activation storage

Training records a private contiguous FP32 tape per sample and head together with one-based INT32 LU pivots. The backward reuses the saved LU factors to apply the transposed equilibrium solve, then evaluates analytic VJPs for the dynamic core, accretive parameterization, compact cross-moment, soft frame, and Rank-Rotary transform. It does not differentiate through an unrolled solver. The token-level frame and relation derivatives are fused, so the single-consumer frame adjoint remains on chip rather than making a global $[BH, N, r]$ round trip. The complement reduction reconstructs its frame-content term in FP32 within an existing compact kernel. Buffers are reused after their last consumer: the compact-core matrix adjoint becomes the frame adjoint, and compact-state partial storage is overwritten after its reduction.

Let $S = BH$, $T_{32} = \lceil N/32 \rceil$, $T_{64} = \lceil N/64 \rceil$, $Q = \lceil d/32 \rceil$, and $C_T = \min(T_{32}, 64)$. Excluding packed input, output, returned gradients, Rank-Rotary phase tables, and allocator bookkeeping, the saved training tape contains

$$W_{\text{train}} = S(2Nr + 3r^2 + 2rd + 1) \quad (45)$$

FP32 values and Sr INT32 pivots. The compact inference workspace contains

$$W_{\text{infer}} = S(Nr + 2r^2 + 2rd + 1) \quad (46)$$

FP32 values, plus Sr transient INT32 pivots. Rank-Rotary phase storage is additional: a shared table contains Nr FP32 values, while per-sample positions require BNr . Default consecutive-position tables live in a bounded persistent cache; explicit position tables are invocation-local.

The long-sequence Gram schedule has the logical live storage

$$W_{\text{Gram}} = ST_{32} \frac{r(r+1)}{2} \quad (47)$$

in FP32. Tape-producing calls with $r \in \{16, 32, 48\}$ place these partials in the not-yet-materialized frame region; inference and $r = 64$ training allocate them separately. The cross-moment producer uses

$$W_{\text{cross}} = SQC_T r \, 32 \quad (48)$$

FP32 values, and its lifetime does not overlap the Gram temporary. Native backward scratch contains

$$W_{\text{backward}} = S(T_{64}rd + T_{64} + 3rd + 2r^2) \quad (49)$$

FP32 values. These formulas describe the principal tape/workspace and the explicitly listed scratch tensors owned by the native mixer; full-module peak memory additionally includes phase storage, feature projections, external autograd state, returned gradients, and allocator caching.

B.4 Numerical and build coverage

The backend follows a fixed mixed TF32/FP16-Tensor-Core/FP32 contract. The joint input projection uses TF32 where supported. Selected compact contractions and the readout use FP16 multiplicands with FP32 accumulation. Rank-Rotary phases, the soft-frame Gram and Cholesky factorization, the one-ULP-interiorized complement, LL^T , LU factorization, and triangular solves remain FP32. This partition is implementation behavior, not a model variant.

FP64 evaluation of the sole reference operator is the numerical oracle. The CUDA acceptance envelope requires finite outputs and gradients, forward relative ℓ_2 error at most 5×10^{-3} , and input and parameter gradient error at most 10^{-2} . The oracle suite covers all four ranks, non-tile-aligned head widths, lengths through 4776, the $N = 4096, D = 256, H = 8, r = 32$ long-sequence setting, parallel Gram reduction, batch-specific positions, gapped masks, entirely masked samples, and agreement between inference and tape-producing forward entry points. These are mixed-precision error envelopes rather than bitwise FP32 claims.

CUDA 12.8 with cuBLASDx and cuSolverDx from MathDx 25.12.1 builds separate artifacts for

$$\text{SM} \in \{75, 80, 86, 87, 89, 90, 100, 120\}. \quad (50)$$

Each loaded binary verifies its compiled SM and native contract version, and release metadata records the PyTorch version, CUDA runtime, C++ ABI, and artifact hashes. This is build and dispatch coverage, not a claim that latency or numerical error has been measured on every listed physical architecture.

B.5 Wall-clock protocol

Table 3 was measured under WSL2 on one NVIDIA GeForce RTX 5070 Ti (16 GB, SM120), driver 610.88, PyTorch 2.11.0+cu128, CUDA 12.8, and native contract version 6. Each process constructs one complete mixer in training mode except for inference measurements, pre-generates a contiguous input tensor, and enables TF32. LSSO uses the default DYNAMIC plus Rank-Rotary configuration with consecutive positions and no padding mask. The comparison is PyTorch `nn.MultiheadAttention` with bias, zero dropout, batch-first layout, and `need_weights=False`; profiler traces confirmed that its scaled-dot-product attention selected PyTorch’s Flash SDPA forward and backward kernels.

The timed boundary includes both operators’ input and output projections, but excludes residual connections, MLPs, data loading, and optimizer updates. The forward-backward measurement differentiates a squared-mean scalar loss and includes setting the input gradient to `None` every iteration and model parameter gradients to `None` after each iteration. No CUDA Graph or `torch.compile` path is enabled. Every configuration runs 50 untimed warmup iterations, followed by seven repetitions of 100 iterations. Each repetition is bracketed by `torch.cuda.synchronize()`, and the reported number is the median synchronized host wall-clock time per iteration. Batch, width, head count, rank, precision, and statistical protocol are fixed across all sequence lengths and implementations. The benchmark entry point is `benchmarks/benchmark.ball.py`.

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